LLM Parallelism Foundations: From One GPU to a Distributed Model

Deep Dive into Kimi K3 — Chapter 33

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Foundation Books Project

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Goal: declare layouts and groups, follow each axis's dependency to the exchange it needs, order and retire buffers safely, and keep cost and K3 claims in scope.

Roadmap

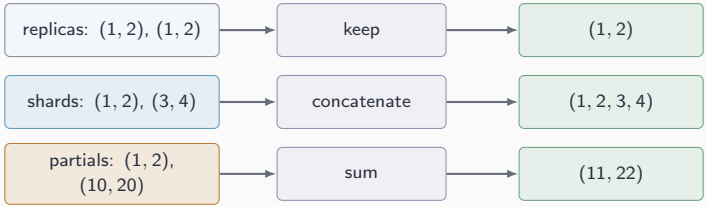
Layouts and collectives

Five axes, five obligations

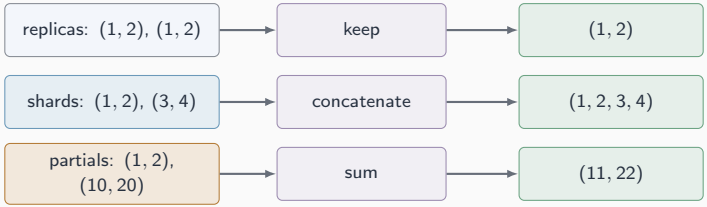
Ordering, composition, and scope

Layouts and collectives

Keep replicas, concatenate shards, sum partials



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- Same arithmetic, different holders: all-reduce gives G to every rank; reduce-scatter leaves $G[I_r]$ at rank r; all-gather after reduce-scatter equals all-reduce. [DERIVED: four-rank fixture]

Five axes, five obligations

What each axis divides, and what it must exchange

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- **CP:** a causal query needs the remote prefix ($o_5 = 3$, not 5); a recurrence needs the incoming state ($s_5 = 15$, not 5).

[DERIVED: constructed CPU witnesses]

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Correctness first, cost as a separate question. Next: Chapter 34, FlashKDA and KDA Kernels.